AI-PRISM (LUAD)

Key hyperparameters ? Current configuration

Hyperparameters are settings chosen before training that shape the model, learning process, or evaluation. Values below reflect config.yaml.

Input and model

Setting	Current value	What it controls
model_name	Simonlee711/Clinical_ ModernBERT	Pretrained text encoder. Training loads this model and fine-tunes some of its layers.
max_len	4096 tokens	Maximum report length given to the tokenizer. Longer reports are truncated; increasing this preserves more text but uses more memory.
pooling	mean	How token embeddings become one report embedding. **mean** averages non-padding tokens; **cls** uses the first token.
head_hidden	256	Width of each gene-specific MLP. With a 768-dimensional encoder, each head is 768 ? 256 ? 128 ? 1.
dropout	0.2	Fraction of activations randomly disabled during training to reduce overfitting. Used in the prediction heads and before them.
grad_ckpt	false	Gradient checkpointing trades extra computation for lower memory use. Enable if training runs out of GPU memory.

Loss and class imbalance

Setting	Current value	What it controls
loss	bce	Training objective: binary cross-entropy (**bce**) or asymmetric loss (**asl**).
pos_weight	false	For BCE, whether to weight positive examples more heavily for rare mutations. When enabled, weights are derived from training-fold prevalence.
gamma_pos / gamma_neg / margin	0.0 / 4.0 / 0.05	ASL-specific controls for down-weighting easy examples and shifting negative probabilities. These have no effect with the current **loss: bce**.

Training

Setting	Current value	What it controls
batch_size	2	Patients processed per training batch.
grad_accum	8	Batches accumulated before an optimizer update. Together with batch_size, effective batch size is 16 patients (2 ? 8), except the final partial accumulation.
eval_batch_size	4	Patients processed per validation or test batch; does not change optimizer updates.
lr_head	0.001	Learning rate for the gene-specific prediction heads.
lr_encoder	0.00002	Lower learning rate for trainable encoder layers during fine-tuning.
phase1_epochs	5	Maximum epochs in phase 1, when the encoder is frozen and only the heads train.
phase2_epochs	15	Maximum epochs in phase 2, when the configured top encoder layers are unfrozen and fine-tuned.
unfreeze_layers	4	Number of top encoder layers made trainable in phase 2.
weight_decay	0.01	AdamW regularization that discourages overly large weights.
warmup	0.1	Fraction of optimizer steps used to ramp the learning rate up before it decays.
max_grad_norm	1.0	Clips large gradients to help stabilize training.
patience	4	Stop a phase after four consecutive epochs without improved validation macro-AUPRC.

Data splitting and reproducibility

Setting	Current value	What it controls
n_folds	5	Number of multilabel-stratified cross-validation folds. Each fold has its own held-out test set.
seed	42	Seeds Python, NumPy, and PyTorch randomness to make splits and training more reproducible.
min_chars	100	Drops reports shorter than 100 characters before splitting and training.

In brief: the current run uses 4096-token reports, a pretrained clinical encoder, and five gene-specific heads. It trains the heads with the encoder frozen, then fine-tunes the top four encoder layers. The checkpoint with the best validation macro-AUPRC is selected.